

# DBMS – SELLER AND INVENTORY MANAGEMENT

**Project Title:** Helios Watch Store E-Commerce Database

**Skill Task:** DBMS

**Week:** 3

## 1. Database Design

### 1.1 Seller Table

| Attribute   | Data Type    | Constraint       |
|-------------|--------------|------------------|
| Seller_ID   | INT          | Primary Key      |
| Seller_Name | VARCHAR(100) | NOT NULL         |
| Email       | VARCHAR(100) | NOT NULL, UNIQUE |
| Contact_No  | VARCHAR(15)  | NOT NULL, UNIQUE |
| Address     | VARCHAR(200) | NOT NULL         |

### 1.2 Inventory Table

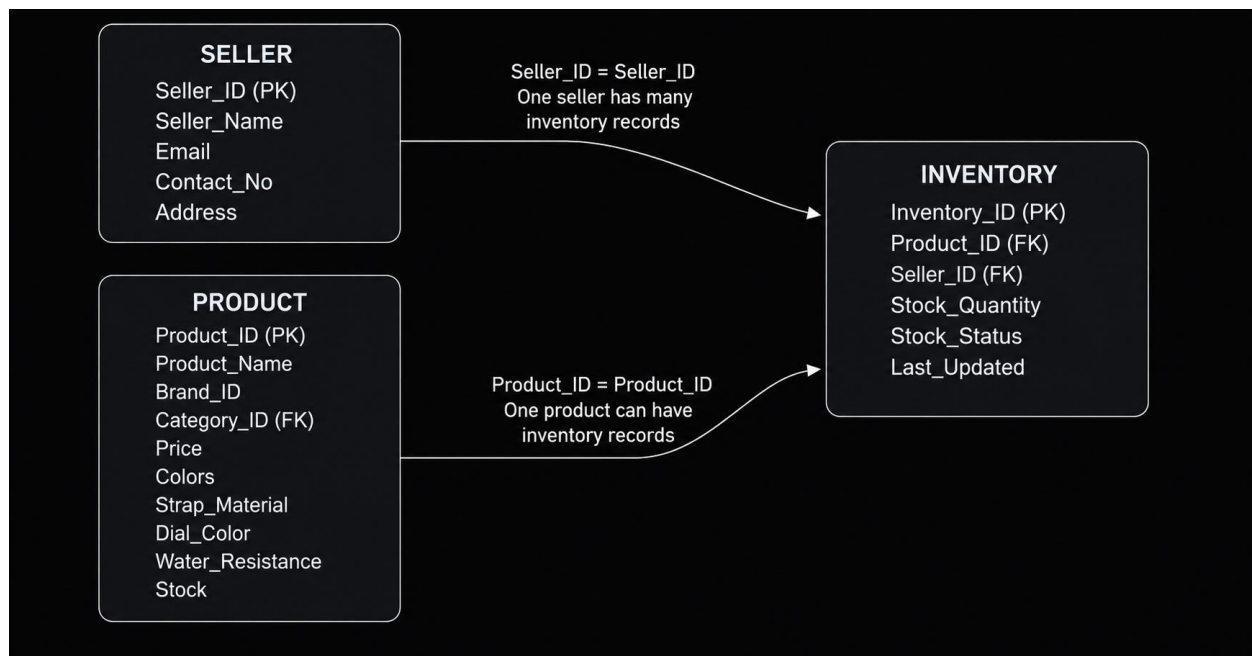
| Attribute      | Data Type   | Constraint  |
|----------------|-------------|-------------|
| Inventory_ID   | INT         | Primary Key |
| Product_ID     | INT         | Foreign Key |
| Seller_ID      | INT         | Foreign Key |
| Stock_Quantity | INT         | NOT NULL    |
| Stock_Status   | VARCHAR(20) | NOT NULL    |
| Last_Updated   | DATE        | NOT NULL    |

## 2. Table Relationships

The **Seller** and **Inventory** tables are related through Seller\_ID.

- **Seller → Inventory:** One seller can have multiple inventory records.
- **Inventory → Seller:** Each inventory record belongs to one seller.
- **Product → Inventory:** One product can have an inventory record.
- **Inventory → Product:** Each inventory record refers to one product.
- Seller\_ID in the **Inventory** table is a **Foreign Key** referencing Seller(Seller\_ID).
- Product\_ID in the **Inventory** table is a **Foreign Key** referencing Product(Product\_ID).

## Relationship Representation



## 3. Table Creation

### 3.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID    NUMBER PRIMARY KEY,  
3     Seller_Name  VARCHAR2(100) NOT NULL,  
4     Email        VARCHAR2(100) NOT NULL UNIQUE,  
5     Contact_No   VARCHAR2(15) NOT NULL UNIQUE,  
6     Address      VARCHAR2(200) NOT NULL  
7 );
```

Table created.

### 3.2 Create Inventory Table

```

SQL> CREATE TABLE Inventory (
  2     Inventory_ID    NUMBER PRIMARY KEY,
  3     Product_ID      NUMBER NOT NULL,
  4     Seller_ID        NUMBER NOT NULL,
  5     Stock_Quantity  NUMBER NOT NULL,
  6     Stock_Status    VARCHAR2(20) NOT NULL,
  7     Last_Updated    DATE NOT NULL,
  8     CONSTRAINT FK_Inventory_Product
  9         FOREIGN KEY (Product_ID)
10         REFERENCES Product(Product_ID),
11     CONSTRAINT FK_Inventory_Seller
12         FOREIGN KEY (Seller_ID)
13         REFERENCES Seller(Seller_ID)
14 );

```

Table created.

#### 4. Seller Data Insertion

```

SQL> INSERT INTO Seller
  2 VALUES (1, 'TimeZone Watches', 'timezone@gmail.com', '9876500001', 'Chennai');

1 row created.

SQL>
SQL> INSERT INTO Seller
  2 VALUES (2, 'Watch World', 'watchworld@gmail.com', '9876500002', 'Coimbatore');

1 row created.

SQL>
SQL> INSERT INTO Seller
  2 VALUES (3, 'Wrist Point', 'wristpoint@gmail.com', '9876500003', 'Madurai');

1 row created.

SQL>
SQL> INSERT INTO Seller
  2 VALUES (4, 'Elite Watches', 'elitewatches@gmail.com', '9876500004', 'Salem');

1 row created.

SQL>
SQL> INSERT INTO Seller
  2 VALUES (5, 'Classic Time', 'classictime@gmail.com', '9876500005', 'Trichy');

1 row created.

```

Activate W

```
SQL>
SQL> INSERT INTO Seller
  2 VALUES (6, 'Smart Wrist', 'smartwrist@gmail.com', '9876500006', 'Erode');

1 row created.

SQL>
SQL> INSERT INTO Seller
  2 VALUES (7, 'Royal Timepieces', 'royaltimepieces@gmail.com', '9876500007', 'Vellore');

1 row created.

SQL>
SQL> INSERT INTO Seller
  2 VALUES (8, 'Watch Hub', 'watchhub@gmail.com', '9876500008', 'Thanjavur');

1 row created.
```

## 5. Inventory Data Insertion

```
SQL> INSERT INTO Inventory
  2 VALUES (1, 101, 1, 25, 'Available', TO_DATE('01-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (2, 102, 2, 18, 'Available', TO_DATE('02-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (3, 103, 3, 30, 'Available', TO_DATE('03-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (4, 104, 4, 15, 'Available', TO_DATE('04-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (5, 105, 5, 12, 'Available', TO_DATE('05-09-2026', 'DD-MM-YYYY'));

1 row created.
```

```

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (6, 106, 6, 20, 'Available', TO_DATE('06-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (7, 107, 7, 0, 'Unavailable', TO_DATE('07-09-2026', 'DD-MM-YYYY'));

1 row created.

SQL>
SQL> INSERT INTO Inventory
  2 VALUES (8, 108, 8, 0, 'Unavailable', TO_DATE('08-09-2026', 'DD-MM-YYYY'));

1 row created.

```

## 6. Seller Product Information

### 6.1 Display Seller and Product Details

This query combines **Seller**, **Inventory**, and **Product** information.

```

SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     p.Product_ID,
  5     p.Product_Name,
  6     i.Stock_Quantity,
  7     i.Stock_Status,
  8     i.Last_Updated
  9 FROM Seller s
 10 JOIN Inventory i
 11     ON s.Seller_ID = i.Seller_ID
 12 JOIN Product p
 13     ON i.Product_ID = p.Product_ID
 14 ORDER BY s.Seller_Name;

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          5
Classic Time
          105

```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
Smart Active Watch
          12 Available          05-SEP-26
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          4
Elite Watches
          104
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
Smart Fit Pro
          15 Available          04-SEP-26
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          7
Royal Timepieces
          107
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
Outdoor Pro Watch
          10 Available          12-SEP-26
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          6
Smart Wrist
        106
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
Sports Chronograph
          20 Available          06-SEP-26
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          1
TimeZone Watches
        101
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
Classic Silver Watch
          25 Available          01-SEP-26
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          8
Watch Hub
          108
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
Royal Gold Edition
          0 Unavailable          08-SEP-26
```

```
SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          2
Watch World
          102
```

```

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
Elegant Gold Watch
          18 Available          02-SEP-26

```

```

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          3
Wrist Point
          103

```

```

SELLER_ID
-----
SELLER_NAME
-----
PRODUCT_ID
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
Digital Sport Watch
          30 Available          03-SEP-26

8 rows selected.

```

## 7. Inventory Status

## 7.1 Display Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM Product p
  8 JOIN Inventory i
  9     ON p.Product_ID = i.Product_ID
 10 JOIN Seller s
 11     ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Status = 'Available';
```

PRODUCT\_ID

-----

PRODUCT\_NAME

-----

SELLER\_NAME

-----

STOCK\_QUANTITY STOCK\_STATUS

-----

101

Classic Silver Watch

TimeZone Watches

25 Available

PRODUCT\_ID

-----

PRODUCT\_NAME

-----

SELLER\_NAME

-----

STOCK\_QUANTITY STOCK\_STATUS

-----

102

Elegant Gold Watch

Watch World

18 Available

PRODUCT\_ID

-----

PRODUCT\_NAME

-----

SELLER\_NAME

-----

STOCK\_QUANTITY STOCK\_STATUS

-----

103

Digital Sport Watch

Wrist Point

30 Available

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
          104
Smart Fit Pro
Elite Watches
          15 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
          105
Smart Active Watch
Classic Time
          12 Available
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
          106
Sports Chronograph
Smart Wrist
          20 Available
```

6 rows selected.

## 7.2 Display Unavailable Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status
  7 FROM Product p
  8 JOIN Inventory i
  9     ON p.Product_ID = i.Product_ID
 10 JOIN Seller s
 11     ON i.Seller_ID = s.Seller_ID
 12 WHERE i.Stock_Status = 'Unavailable';
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
          107
Outdoor Pro Watch
Royal Timepieces
          0 Unavailable
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
          108
Royal Gold Edition
Watch Hub
          0 Unavailable
```

## 8. Inventory Update

Demonstrate updating stock information.

```
SQL>
SQL> UPDATE Inventory
  2 SET Stock_Quantity = 10,
  3     Stock_Status = 'Available',
  4     Last_Updated = TO_DATE('12-09-2026', 'DD-MM-YYYY')
  5 WHERE Product_ID = 107;

1 row updated.
```

## 9. Inventory Status Reports

### 9.1 Complete Inventory Report

```
SQL> SELECT
  2     i.Inventory_ID,
  3     s.Seller_Name,
  4     p.Product_Name,
  5     i.Stock_Quantity,
  6     i.Stock_Status,
  7     i.Last_Updated
  8 FROM Inventory i
  9 JOIN Seller s
 10     ON i.Seller_ID = s.Seller_ID
 11 JOIN Product p
 12     ON i.Product_ID = p.Product_ID
 13 ORDER BY i.Inventory_ID;
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
1
TimeZone Watches
Classic Silver Watch
      25 Available      01-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
2
Watch World
Elegant Gold Watch
      18 Available      02-SEP-26
```

```
INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
3
Wrist Point
Digital Sport Watch
      30 Available      03-SEP-26
```

```

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          4
Elite Watches
Smart Fit Pro
          15 Available      04-SEP-26

```

```

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          5
Classic Time
Smart Active Watch
          12 Available      05-SEP-26

```

```

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          6
Smart Wrist
Sports Chronograph
          20 Available      06-SEP-26

```

```

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
          7
Royal Timepieces
Outdoor Pro Watch
          10 Available      12-SEP-26

```

```

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME
-----
STOCK_QUANTITY STOCK_STATUS      LAST_UPDA
-----
            8
Watch Hub
Royal Gold Edition
            0 Unavailable      08-SEP-26

8 rows selected.

```

## 9.2 Available and Unavailable Product Count

```

SQL> SELECT
      2      Stock_Status,
      3      COUNT(*) AS Product_Count
      4  FROM Inventory
      5  GROUP BY Stock_Status;

STOCK_STATUS      PRODUCT_COUNT
-----
Available              7
Unavailable           1

```

### 9.3 Total Stock by Seller

```
SQL> SELECT
  2     s.Seller_Name,
  3     SUM(i.Stock_Quantity) AS Total_Stock
  4 FROM Seller s
  5 JOIN Inventory i
  6     ON s.Seller_ID = i.Seller_ID
  7 GROUP BY s.Seller_Name
  8 ORDER BY s.Seller_Name;
```

SELLER\_NAME

TOTAL\_STOCK

Classic Time  
12

Elite Watches  
15

Royal Timepieces  
10

SELLER\_NAME

TOTAL\_STOCK

Smart Wrist  
20

TimeZone Watches  
25

Watch Hub  
0

SELLER\_NAME

TOTAL\_STOCK

Watch World  
18

Wrist Point  
30

8 rows selected.

## 9.4 Products with Zero Stock

This is particularly useful for demonstrating unavailable products.

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     s.Seller_Name,
  5     i.Stock_Quantity
  6 FROM Product p
  7 JOIN Inventory i
  8     ON p.Product_ID = i.Product_ID
  9 JOIN Seller s
 10     ON i.Seller_ID = s.Seller_ID
 11 WHERE i.Stock_Quantity = 0;
```

PRODUCT\_ID

-----

PRODUCT\_NAME

-----

SELLER\_NAME

-----

STOCK\_QUANTITY

-----

108

Royal Gold Edition

Watch Hub

0